Performance Testing

Date	2 July 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Step 1: Testing Overview	
Testing Tool Used	Postman + Browser-based Manual Testing + Flask Debug Server
Type of Testing	Functional, Load (manual), Response Time, Accuracy
Target Module/API	/predict endpoint (POST)
Test Environment	Local Flask Server (http://127.0.0.1:5000)
Step 2: Test Scenarios	
1	Home page loads successfully (<2 sec)
2	User login with valid credentials
3	Crop prediction using valid soil/environment values
4	Multiple consecutive prediction requests
5	Invalid input validation
Step 3: Performance Results	
Avg Response Time	~0.9 sec (Pass)
Max Response Time	~1.4 sec (Pass)
Throughput	>7 requests/sec (Pass)
Error Rate	0% (Pass)
CPU Utilization	~18% (Pass)
Memory Utilization	~14% (Pass)

Step 4: Observations	
Analysis	The Smart Agriculture Flask application performed e during local testing. T